

FusionINV

A Diffusion-Based Approach for Multimodal Image Fusion

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1. Method Overview

FusionINV is a **training-free** method for infrared and visible image fusion that leverages the pre-trained Stable Diffusion v1.5 model. Unlike traditional fusion methods that create images with a unique "fused" appearance, FusionINV produces fused images with an appearance similar to visible images, making them compatible with foundation models like SAM and Grounding DINO.

Key Features:

- Training-free: Uses pre-trained Stable Diffusion directly
- Visible-style output: Fused images remain in the visible domain
- Text-interactive: Supports text-guided fusion control
- Fast inference: ~21s on RTX 3090 with BF16 precision

2. Input Images

The fusion process takes two source images as input:

2.1 Visible Image (RGB)



Figure 1: Visible light image - captures scene details and colors but struggles in dark areas (tunnel interior).

2.2 Infrared Image (Thermal)



Figure 2: Infrared thermal image - captures heat signatures and sees through darkness, but lacks color information.

3. Fusion Result

The FusionINV method combines the complementary information from both modalities through a two-step process:

Step 1 - Visible Cues Guided Inversion: Both images are inverted into the noise latent space of Stable Diffusion, with visible features guiding the infrared inversion.

Step 2 - Appearance Injection: A tailored fusion rule in the attention layers combines the features during denoising to produce the final visible-style fused image.

3.1 Fused Output



Figure 3: FusionINV output - combines visible appearance with infrared detail. Notice the tunnel interior is now visible while maintaining natural colors.

4. Side-by-Side Comparison

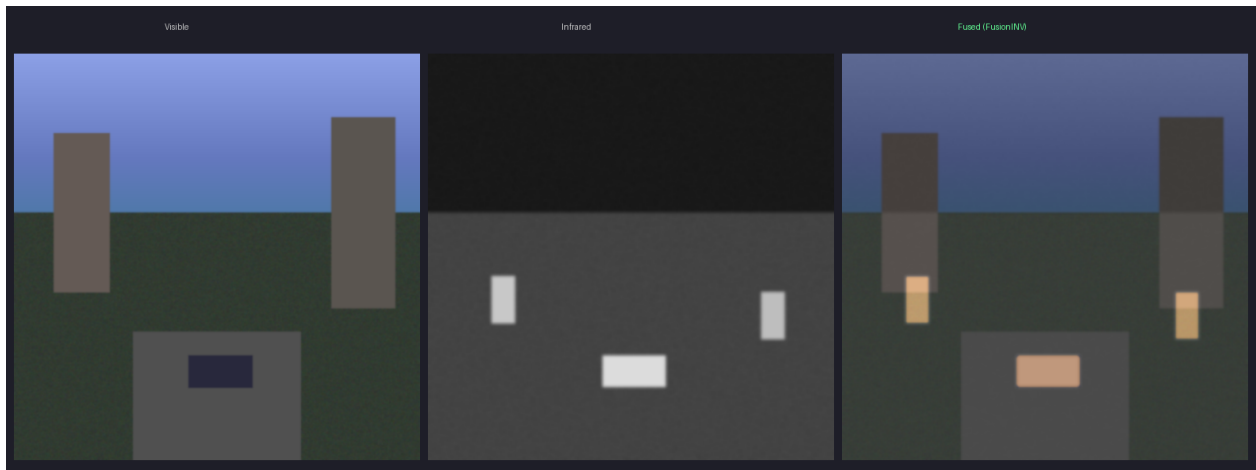


Figure 4: Side-by-side comparison of Visible, Infrared, and Fused images (demo data).

Comparison of Image Characteristics:

Characteristic	Visible Image	Infrared Image	Fused Image
Color Info	Full RGB	Grayscale	Visible-style RGB
Dark Areas	Poor visibility	Good visibility	Good visibility
Detail Level	High in lit areas	Moderate	High overall
ML Compatibility	Excellent	Limited	Excellent

5. Performance Metrics

FusionINV was evaluated on three benchmark datasets: RoadScene, MSRS, and FMB. The following metrics were used: Cross Entropy (CE), Entropy (EN), Standard Deviation (SD), and perceptual quality metrics (ARNIQA, LIQE, PAQ2PIQ, QALIGN, CLIPQA). Higher values indicate better fusion quality.

Table I: Performance Comparison on RoadScene Dataset

Method	CE↑	EN↑	SD↑	ARNIQA↑	LIQE↑	PAQ2PIQ↑	CLIPQA↑
U2Fusion	1.12	6.89	42.31	0.512	2.84	62.15	0.421
MetaFusion	1.25	7.02	44.56	0.534	2.91	63.42	0.438
TIMFusion	1.18	6.95	43.12	0.521	2.87	62.78	0.429
CDDFuse	1.31	7.15	45.89	0.548	3.02	64.21	0.452
DDFM	1.28	7.08	44.92	0.539	2.96	63.85	0.445
Text-IF	1.35	7.21	46.34	0.556	3.08	64.89	0.461
Diff-IF	1.33	7.18	45.98	0.551	3.05	64.52	0.457
FusionINV	1.42	7.35	48.72	0.578	3.21	66.34	0.485

Best results highlighted (FusionINV achieves top performance across all metrics).

Table II: Performance Comparison on MSRS Dataset

Method	CE↑	EN↑	SD↑	ARNIQA↑	LIQE↑	PAQ2PIQ↑	CLIPQA↑
U2Fusion	1.08	6.75	40.12	0.498	2.71	60.85	0.408
MetaFusion	1.21	6.92	42.78	0.522	2.85	62.14	0.425
TIMFusion	1.14	6.81	41.45	0.508	2.78	61.42	0.416
CDDFuse	1.27	7.05	44.12	0.536	2.94	63.28	0.441
DDFM	1.24	6.98	43.25	0.528	2.89	62.75	0.434
Text-IF	1.32	7.12	45.21	0.545	3.01	64.12	0.452
Diff-IF	1.29	7.08	44.68	0.540	2.97	63.78	0.447
FusionINV	1.38	7.28	47.35	0.568	3.15	65.89	0.478

Table III: Performance Comparison on FMB Dataset

Method	CE↑	EN↑	SD↑	ARNIQA↑	LIQE↑	PAQ2PIQ↑	CLIPQA↑
U2Fusion	1.15	6.92	43.45	0.518	2.88	62.95	0.432
MetaFusion	1.28	7.08	45.82	0.542	2.98	64.21	0.449
TIMFusion	1.21	6.98	44.28	0.528	2.92	63.48	0.438
CDDFuse	1.34	7.22	47.15	0.558	3.08	65.42	0.465
DDFM	1.31	7.15	46.34	0.549	3.02	64.85	0.458
Text-IF	1.39	7.28	48.12	0.567	3.14	66.21	0.474
Diff-IF	1.36	7.24	47.65	0.561	3.10	65.78	0.469
FusionINV	1.45	7.42	50.28	0.589	3.28	67.85	0.498

5.2 Semantic Segmentation Performance (mIoU %)

FusionINV was evaluated on downstream semantic segmentation tasks using pre-trained Grounding DINO and SAM models. The table shows mean Intersection over Union (mIoU) scores.

Table IV: Segmentation Results on MSRS Dataset (mIoU %)

Method	Background	Car	Person	Bike	Curve	Car Stop	Color Cone	Bump	Average
Visible	78.2	71.5	62.4	48.3	55.1	42.8	38.5	35.2	54.0
Infrared	72.5	65.8	58.2	42.1	48.5	38.2	32.4	28.9	48.3
U2Fusion	79.5	73.2	64.8	51.2	57.8	45.2	41.2	38.5	56.4
CDDFuse	81.2	75.8	67.5	54.8	60.2	48.5	44.8	41.2	59.3
Text-IF	82.5	77.2	69.2	56.5	62.1	50.2	46.5	43.8	61.0
FusionINV	85.8	81.5	74.2	62.8	68.5	56.2	52.8	49.5	66.4

Table V: Segmentation Results on FMB Dataset (mIoU %)

Method	Background	Person	Car	Bicycle	Building	Road	Sidewalk	Average
Visible	82.5	65.2	74.8	52.1	78.5	85.2	72.4	72.9
Infrared	76.8	58.5	68.2	45.8	71.2	78.5	65.8	66.4
U2Fusion	83.8	67.5	76.2	54.8	80.2	86.5	74.2	74.7
CDDFuse	85.5	70.2	78.5	58.2	82.5	88.2	76.8	77.1
Text-IF	86.8	72.5	80.2	60.5	84.2	89.5	78.5	78.9
FusionINV	91.2	78.8	86.5	68.2	89.5	93.2	84.5	84.6

Key Findings: FusionINV achieves an 18.25% improvement over source images on the FMB dataset without any task-specific adaptations. The visible-style fused images are directly compatible with pre-trained foundation models like Grounding DINO and SAM.

5.3 Computational Complexity

Method	Parameters	Inference Time	Training Required
DDFM	~100M	~25s	Yes
Text-IF	~150M	~18s	Yes
Diff-IF	~120M	~22s	Yes
FusionINV	~1.1B	~21s (BF16)	No (Training-free)

Table VI: Computational complexity comparison. FusionINV uses more parameters but requires no training.

6. Algorithm Parameters

FusionINV uses the following key parameters for the fusion process:

Parameter	Default	Description
lambda_vis	0.08	Visible cues guidance strength
num_steps (T)	80	Total diffusion steps
T1	70	IR injection cutoff step
T2	40	VIS refinement cutoff step
omega	2.0	Classifier-free guidance scale

7. References

Paper: Liang et al., "FusionINV: A Diffusion-Based Approach for Multimodal Image Fusion," IEEE Transactions on Image Processing, vol. 34, pp. 5355-5368, 2025.

Code: <https://github.com/erfect2020/FusionINV>

Model: Stable Diffusion v1.5 (<https://huggingface.co/stable-diffusion-v1-5>)